

LICENCIATURA EN CIENCIA DE DATOS

COD DE MATERIA: 177

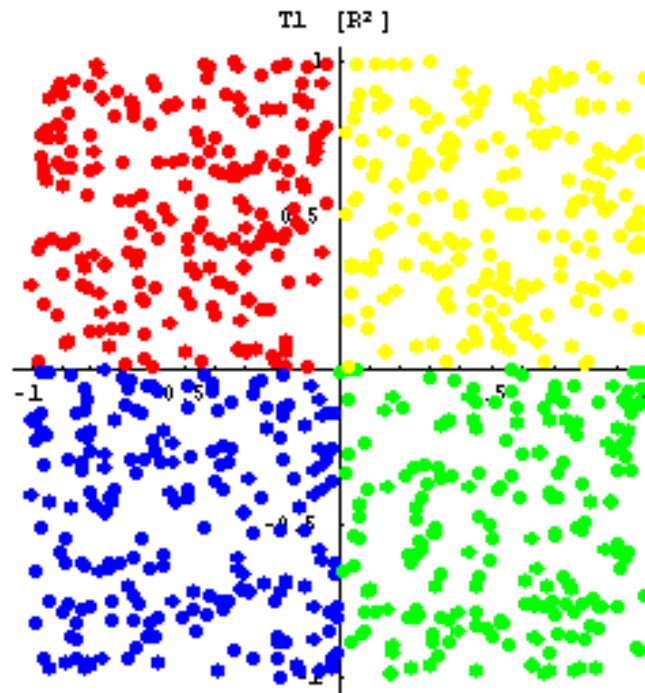
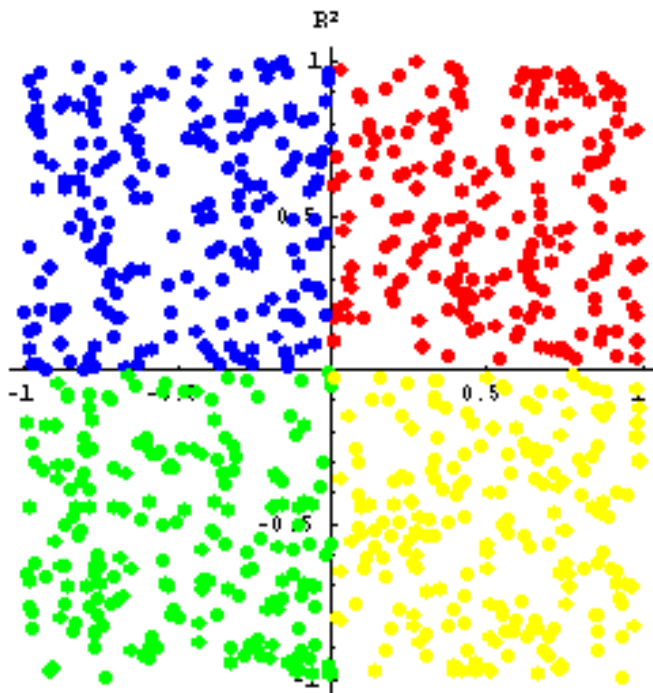
JTP: ING. GUIDO TRUCCO

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Interpretación gráfica de las Transformaciones lineales

Sea $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \mid T(x, y) = (-y, x)$



$$(1, 2) \quad T(1, 2) = (-2, 1)$$

$$(-1, 2) \quad T(-1, 2) = (-2, -1)$$

$$(3, -2) \quad T(3, -2) = (2, 3)$$

Interpretación gráfica de las Transformaciones lineales

Sea $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 | T(\mathbf{x}, \mathbf{y}) = (-\mathbf{y}, \mathbf{x})$

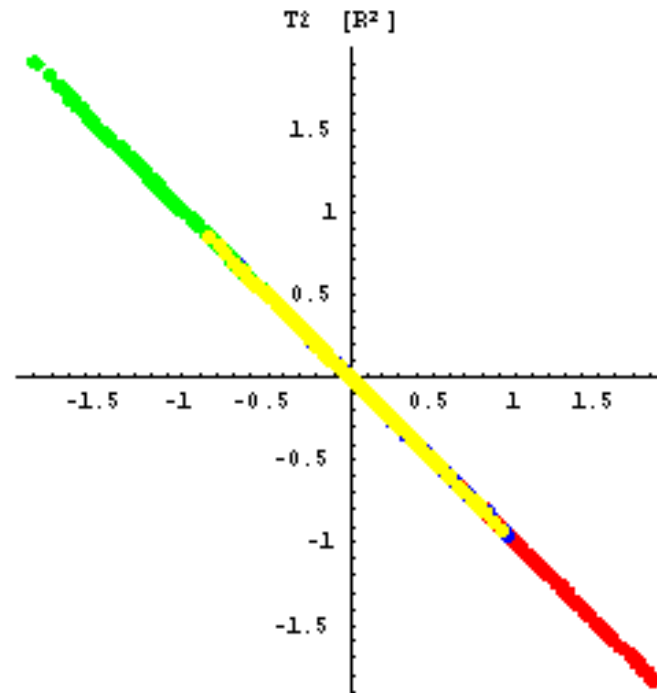
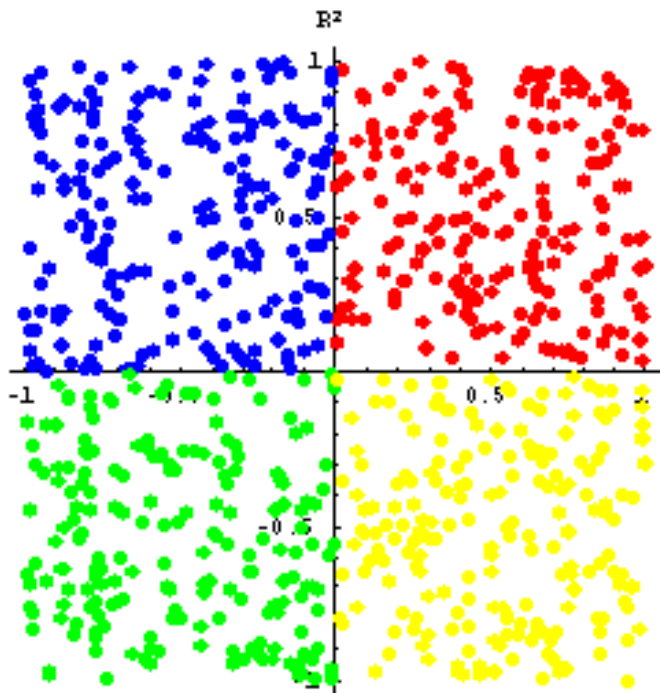
Cálculo analítico de la imagen:

$$\text{Img}(T) = \{\mathbf{y} \in \mathbb{R}^2 | \forall \mathbf{x} \in \mathbb{R}^2 \quad T(\mathbf{x}) = \mathbf{y}\}$$

Si $\mathbf{y} = (y_1, y_2) = T(x_1, x_2) = (-x_2, x_1)$ entonces
$$\begin{cases} y_1 = -x_2 \\ y_2 = x_1 \end{cases}$$

Interpretación gráfica de las Transformaciones lineales

Sea $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \mid T(x, y) = (x + y, -x - y)$



- $(1, 1) \quad T(1, 1) = (2, -2)$
- $(-1, 2) \quad T(-1, 2) = (3, -3)$
- $(3, -2) \quad T(3, -2) = (1, -1)$

Interpretación gráfica de las Transformaciones lineales

Sea $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \mid T(x, y) = (x + y, -x - y)$

$$\text{Img}(T) = \{\mathbf{y} \in \mathbb{R}^2 \mid \forall \mathbf{x} \in \mathbb{R}^2 \quad T(\mathbf{x}) = \mathbf{y}\}$$

Si $\mathbf{y} = (y_1, y_2) = T(x_1, x_2) = (x_1 + x_2, -x_1 - x_2)$

Entonces $x_1 + x_2 = y_1$; $-x_1 - x_2 = y_2$ Así $y_1 = -y_2$

$$\text{Img}(T) = \{\mathbf{y} \in \mathbb{R}^2 \mid y_1 = -y_2\}$$